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Education

- University of Waterloo**, Waterloo, ON Sep 2019 – Sep 2025
Ph.D. in Computer Science – Advisor: Chengnian Sun
- Northeastern University**, Boston, MA Sep 2016 – Dec 2018
B.S. in Computer Science – Minor in Computer Engineering (GPA: 3.9/4.0)

Experience (Selected)

- Visiting Researcher**, *University of Waterloo* Sep 2025 – Present
Research Assistant, *University of Waterloo* Fall 2019 – Fall 2025

- Evaluating LLMs for Type Inference on Java Code Snippets** 2024 – Present
- Designed *novel evaluation* framework for LLM-based type inference on code snippets.
 - Identified critical *data leakage* issue in prior evaluations of LLM-based type inference.

- Scalable Type inference on Java Code Snippets** 2022 – Present
- Designed iterative optimizing algorithm to improve type inference scalability over large databases.
 - Optimized SQL database queries with tables containing *millions* of Java types and methods.
 - Improved running time performance by over 20x, while maintaining high accuracy.

- Automated Type Inference on Java Code Snippets** 2020 – 2022
- Developed a novel type inference method, leveraging semantic information in code structures.
 - Engineered static analysis tool using Eclipse JDT to extract relevant type information.
 - Published SnR, *doubling* performance over prior tools by *guaranteeing* type-compatible solution.

- Bash Empirical Study** 2019 – 2021
- Published large-scale study by scraping and processing >1 million Bash scripts from GitHub.
 - Examined the common usage, code smell, and bugs in Bash scripts.
 - Conducted manual analysis of bug reports to classify common bug types.

- Teaching Assistant**, *University of Waterloo* Fall 2019 – Spring 2025
- Held weekly standup to verify project progression and to ensure stakeholder goals are met.
 - Provided feedback to students through periodic meetings, office hours, and markings.
 - Courses: Sequential Programs (CS241), App Dev (CS346), Software Design (CS446), Software Testing (SE465), OOP (CS246), FP (CS135)

Technical Knowledge

Languages

- Java, Python, JavaScript, MySQL, C, C++, Bash

Frameworks

- ANTLR, Spring, Hibernate, React, jQuery, node.js, JSP, JUnit, Maven

Publication (Selected)

- Scitix: Scalable Constraint-Based Type Inference for Code Snippets with Unknown Types. * Revision
Evaluating the Genuine Type Inference Capabilities of LLMs for Java Code Snippets. * Revision
SnR: Constraint-Based Type Inference for Incomplete Java Code Snippets. * ICSE 2022
Bash in the Wild: Language Usage, Code Smells, and Bugs. * TOSEM 2022
LPR: Large Language Models-Aided Program Reduction. ISSTA 2024
Revisiting the Evaluation of Deep Learning-Based Compiler Testing. IJCAI 2023